

Django Machine Learning Lab

Manual/30marks

Building a Vehicle Analytics System with Regression, Classification, and Client Segmentation
Designed as a step-by-step tutorial for students learning Python/Django and Machine Learning integration.

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PART I: Learning Objectives

By the end of this lab students should be able to:

- Create and configure a Django project
- Prepare a dataset for machine learning
- Train machine learning models using scikit-learn
- Implement three ML techniques:
 - ✓ Regression (predict selling price)
 - ✓ Classification (predict income level)
 - ✓ Clustering (client segmentation)
- Integrate ML models into a Django web application
- Build a web interface to interact with ML predictions
- Deploy and test the ML system

PART II: Project Overview

In this project we build a Vehicle Analytics System. The system will analyze a dataset of vehicle sales and provide:

1. Price Prediction: Predict the selling price of a vehicle using regression.
2. Income Classification: Predict the customer's income level using classification.
3. Client Segmentation: Group customers into clusters using K-Means clustering.

1. Technology Stack:

- Django
- Python
- Pandas
- Scikit-learn
- HTML templates

2. Required Software

Install the following tools:

- 🔧 Python 3.9 or newer
- 🔧 Visual Studio Code (recommended)
- 🔧 Git (optional)
- 🔧 Google Chrome or Firefox

Python packages required:

```
django
pandas
scikit-learn
matplotlib
seaborn
joblib
```

PART III: Creating project

3. Create Project Folder

Open a terminal and run:

```
mkdir django_ml_project
cd django_ml_project
```

4. Create Virtual Environment

Create a Python virtual environment.

```
python -m venv venv
```

Activate environment

i. Windows:

```
venv\Scripts\activate
```

ii. Mac/Linux:

```
source venv/bin/activate
```

5. Install Required Libraries

Install project dependencies using requirements.txt

Create a file named `requirements.txt` and add:

```
django
pandas
scikit-learn
matplotlib
seaborn
joblib
plotly
```

Or Generate automatically from your environment:

```
pip freeze > requirements.txt
```

Install all packages at once:

```
pip install -r requirements.txt
```

6. Create Django Project and App

```
django-admin startproject config .
python manage.py startapp predictor
```

Project structure becomes:

```
django_ml_project/
  manage.py
  config/
  predictor/
  requirements.txt
```

7. Register the Application

Open settings.py and add the new app 'predictor' on existing apps:

```
INSTALLED_APPS = [
    #NB: leave existing apps (DO NOT DELETE)
    'predictor',
]
```

8. Add the Dataset

Within your project folder, create another folder named dummy-data for dataset of your project.

```
django_ml_project/
  dummy-data/
    vehicles_ml_dataset.csv
```

PART IV: Data Exploration Using Pandas

Create a file: predictor/data_eploration.py

```
import pandas as pd
# Data Exploration
def dataset_exploration(df):
    table_html = df.head().to_html(
        classes="table table-bordered table-striped table-sm",
        float_format="%.2f",
        justify="center",
        index=False,
    )
    return table_html

# Data description
def data_exploration(df):
    table_html = df.head().to_html(
        classes="table table-bordered table-striped table-sm",
        float_format="%.2f",
        justify="center",
    )
    return table_html
```

Add bellow in predictor/views.py

```
import pandas as pd
from django.shortcuts import render
from predictor.data_exploration import dataset_exploration, data_exploration

def data_exploration_view(request):
    df = pd.read_csv("dummy-data/vehicles_ml_dataset.csv")

    context = {
        "data_exploration": data_exploration(df),
        "dataset_exploration": dataset_exploration(df),
    }

    return render(request, "predictor/index.html", context)

def regression_analysis(request):
    context = {
        "evaluations": evaluate_regression_model()
    }

    if request.method == "POST":
        year = int(request.POST["year"])
        km = float(request.POST["km"])
        seats = int(request.POST["seats"])
        income = float(request.POST["income"])

        prediction = regression_model.predict([[year, km, seats, income]])[0]
        context["price"] = prediction

    return render(request, "predictor/regression_analysis.html", context)
```

Create file: predictor/templates/predictor/index.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Vehicle ML Dashboard</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"
rel="stylesheet">
</head>

<body class="bg-light">
  <div class="container-fluid">
    <div class="row">
      <nav id="sidebarMenu" class="col-md-3 col-lg-2 d-md-block bg-white border-end sticky-top
vh-100 overflow-y-auto p-3">
        <div class="mb-4 ps-2">
          <h5 class="text-primary fw-bold mb-0">Vehicle ML</h5>
          <small class="text-muted">EDA</small>
        </div>

        <div id="data_exploration-nav" class="nav nav-pills flex-column">
          <a class="nav-link mb-1 active" href="#dataset_exploration">Data Exploration</a>
          <a class="nav-link mb-1" href="#data_exploration">Data Description</a>
          <a class="nav-link mb-1" href="{% url 'regression_analysis' %}">Regression
Analysis</a>
          <a class="nav-link mb-1" href="{% url 'classification_analysis'
%}">Classification Analysis</a>
          <a class="nav-link mb-1" href="{% url 'clustering_analysis' %}">Clustering
Analysis</a>
        </div>
      </nav>

      <main class="col-md-9 ms-sm-auto col-lg-10 px-md-4 py-4"
data-bs-spy="scroll"
data-bs-target="#data_exploration-nav"
data-bs-smooth-scroll="true"
tabindex="0">

        <header class="mb-4 border-bottom pb-3">
          <h1 class="fw-bold text-dark">Vehicle Insights Dashboard</h1>
          <p class="text-secondary mb-0">Exploratory Data Analysis</p>
        </header>

        <section id="dataset_exploration" class="mb-5" style="padding-top: 20px;">
          <div class="card shadow-sm border-0">
            <div class="card-header bg-white border-bottom py-3">
              <span class="fw-bold">Data Exploration</span>
            </div>
            <div class="card-body overflow-x-auto">
              {{ dataset_exploration|safe }}
            </div>
          </div>
        </section>

        <section id="data_exploration" class="mb-5" style="padding-top: 20px;">
          <div class="card shadow-sm border-0">
            <div class="card-header bg-white border-bottom py-3">
              <span class="fw-bold">Statistical Analysis</span>
            </div>
            <div class="card-body overflow-x-auto">
              {{ data_exploration|safe }}
            </div>
          </div>
        </section>

      </main>
    </div>
  </div>

  <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

Create file and add below code: predictor/urls.py

```
from django.urls import path
from predictor import views
urlpatterns = [
    path("data_exploration", views.data_exploration_view, name="data_exploration"),]
```

Within config/urls.py add below code:

```
from django.urls import path, include

urlpatterns = [
    path("", include("predictor.urls")),
]
```

9. Running the Server

Run the Django development server.

```
python manage.py runserver
```

Part V: Model training

10. Feature Selection

Select features useful for machine learning. Example features: year, kilometers_driven, seating_capacity, estimated_income,

Target variables: *selling_price* for regression analysis and prediction, *income_level* for classification. For clustering, it is an unsupervised learning no target variable is needed

11. Regression Model – Predict Selling Price

Create file within the project folder: model_generators/regression/train_regression.py

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score
from sklearn.ensemble import RandomForestRegressor
import joblib

df = pd.read_csv("dummy-data/vehicles_ml_dataset.csv")

features = ["year", "kilometers_driven", "seating_capacity", "estimated_income"]
target = "selling_price"

X = df[features]
y = df[target]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Train model
model = RandomForestRegressor(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

# Save model
joblib.dump(model, "regression_model.pkl")

# Predict
predictions = model.predict(X_test)

# Calculate R2 Score
r2 = round(r2_score(y_test, predictions) * 100, 2)

# Create a Comparison DataFrame for the data_exploration
comparison_df = pd.DataFrame(
    {
        "Actual": y_test.values,
        "Predicted": predictions.round(2),
        "Difference": (y_test.values - predictions).round(2),
    }
)

def evaluate_regression_model():
    return {
        "r2": r2,
        "comparison": comparison_df.head(10).to_html(
            classes="table table-bordered table-striped table-sm",
            float_format="%.2f",
            justify="center",
        ),
    }
```

12. Classification Model – Income Level

Create file within the project folder: `model_generators/classification/train_classifier.py`

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
from sklearn.ensemble import RandomForestClassifier
import joblib

df = pd.read_csv("dummy-data/vehicles_ml_dataset.csv")

# Define features and target (target moved up for logical flow)
features = ["year", "kilometers_driven", "seating_capacity", "estimated_income"]
target = "income_level"

X = df[features]
y = df[target]

# Split data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Train model
model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

# Save model
joblib.dump(model, "classification_model.pkl")

# Predict
predictions = model.predict(X_test)

# Calculate Accuracy Score (equivalent to R2 in your regression example)
accuracy = round(accuracy_score(y_test, predictions) * 100, 2)

# Create a Comparison DataFrame for the data_exploration
comparison_df = pd.DataFrame(
    {
        "Actual": y_test.values,
        "Predicted": predictions,
        "Match": y_test.values == predictions,
    }
)

def evaluate_classification_model():
    return {
        "accuracy": accuracy,
        "comparison": comparison_df.head(10).to_html(
            classes="table table-bordered table-striped table-sm",
            justify="center",
        ),
    }
```

13. Clustering Model

Create file within the project folder: `model_generators/clustering/train_cluster.py`

```
import pandas as pd
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
import joblib

SEGMENT_FEATURES = ["estimated_income", "selling_price"]
df = pd.read_csv("dummy-data/vehicles_ml_dataset.csv")
X = df[SEGMENT_FEATURES]

kmeans = KMeans(n_clusters=3, random_state=42, n_init="auto")
df["cluster_id"] = kmeans.fit_predict(X)
centers = kmeans.cluster_centers_
# Sort clusters by income
sorted_clusters = centers[:, 0].argsort()

cluster_mapping = {
    sorted_clusters[0]: "Economy",
    sorted_clusters[1]: "Standard",
    sorted_clusters[2]: "Premium",
}

df["client_class"] = df["cluster_id"].map(cluster_mapping)

joblib.dump(kmeans, "model_generators/clustering/clustering_model.pkl")
silhouette_avg = round(silhouette_score(X, df["cluster_id"]), 2)

cluster_summary = df.groupby("client_class")[SEGMENT_FEATURES].mean()
cluster_counts = df["client_class"].value_counts().reset_index()
cluster_counts.columns = ["client_class", "count"]
cluster_summary = cluster_summary.merge(cluster_counts, on="client_class")
comparison_df = df[["client_name", "estimated_income", "selling_price", "client_class"]]

def evaluate_clustering_model():
    return {
        "silhouette": silhouette_avg,
        "summary": cluster_summary.to_html(
            classes="table table-bordered table-striped table-sm",
            float_format="%.2f",
            justify="center",
            index=False,
        ),
        "comparison": comparison_df.head(10).to_html(
            classes="table table-bordered table-striped table-sm",
            float_format="%.2f",
            justify="center",
            index=False,
        ),
    }
```

14. Update the views.py

```
import joblib
from model_generators.clustering.train_cluster import evaluate_clustering_model
from model_generators.classification.train_classifier import evaluate_classification_model
from model_generators.regression.train_regression import evaluate_regression_model

# Load models once
regression_model = joblib.load("model_generators/regression/regression_model.pkl")
classification_model = joblib.load("model_generators/classification/classification_model.pkl")
clustering_model = joblib.load("model_generators/clustering/clustering_model.pkl")

def classification_analysis(request):
    context = {
        "evaluations": evaluate_classification_model()
    }
    if request.method == "POST":
        year = int(request.POST["year"])
        km = float(request.POST["km"])
        seats = int(request.POST["seats"])
        income = float(request.POST["income"])
        prediction = classification_model.predict([[year, km, seats, income]])[0]

        context["prediction"] = prediction
    return render(request, "predictor/classification_analysis.html", context)

def clustering_analysis(request):
    context = {
        "evaluations": evaluate_clustering_model()
    }
    if request.method == "POST":
        try:
            year = int(request.POST["year"])
            km = float(request.POST["km"])
            seats = int(request.POST["seats"])
            income = float(request.POST["income"])
            # Step 1: Predict price
            predicted_price = regression_model.predict([[year, km, seats, income]])[0]
            # Step 2: Predict cluster
            cluster_id = clustering_model.predict([[income, predicted_price]])[0]
            mapping = {
                0: "Economy",
                1: "Standard",
                2: "Premium"
            }
            context.update({
                "prediction": mapping.get(cluster_id, "Unknown"),
                "price": predicted_price
            })
        except Exception as e:
            context["error"] = str(e)

    return render(request, "predictor/clustering_analysis.html", context)
```

15. Create templates for the project

Create file within the app folder: templates/predictor/index.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Vehicle ML Dashboard</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
</head>

<body class="bg-light">

  <div class="container-fluid">
    <div class="row">

      <nav id="sidebarMenu" class="col-md-3 col-lg-2 d-md-block bg-white border-end sticky-top vh-100 overflow-y-auto p-3">
        <div class="mb-4 ps-2">
          <h5 class="text-primary fw-bold mb-0">Vehicle ML</h5>
          <small class="text-muted">EDA</small>
        </div>

        <div id="data_exploration-nav" class="nav nav-pills flex-column">
          <a class="nav-link mb-1 active" href="#dataset_exploration">Data Exploration</a>
          <a class="nav-link mb-1" href="#data_exploration">Data Description</a>
          <a class="nav-link mb-1" href="{% url 'regression_analysis' %}">Regression Analysis</a>
          <a class="nav-link mb-1" href="{% url 'classification_analysis' %}">Classification Analysis</a>
          <a class="nav-link mb-1" href="{% url 'clustering_analysis' %}">Clustering Analysis</a>
        </div>
      </nav>

      <main class="col-md-9 ms-sm-auto col-lg-10 px-md-4 py-4">
        data-bs-spy="scroll"
        data-bs-target="#data_exploration-nav"
        data-bs-smooth-scroll="true"
        tabindex="0">

        <header class="mb-4 border-bottom pb-3">
          <h1 class="fw-bold text-dark">Vehicle Insights Dashboard</h1>
          <p class="text-secondary mb-0">Exploratory Data Analysis</p>
        </header>

        <section id="dataset_exploration" class="mb-5" style="padding-top: 20px;">
          <div class="card shadow-sm border-0">
            <div class="card-header bg-white border-bottom py-3">
              <span class="fw-bold">Data Exploration</span>
            </div>
            <div class="card-body overflow-x-auto">
              {{ dataset_exploration|safe }}
            </div>
          </div>
        </section>

        <section id="data_exploration" class="mb-5" style="padding-top: 20px;">
          <div class="card shadow-sm border-0">
            <div class="card-header bg-white border-bottom py-3">
              <span class="fw-bold">Statistical Analysis</span>
            </div>
            <div class="card-body overflow-x-auto">
              {{ data_exploration|safe }}
            </div>
          </div>
        </section>

      </main>
    </div>
  </div>

  <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

Create file within the app folder: templates/predictor/regreton_analysis.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Vehicle ML Dashboard</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
</head>

<body class="bg-light">
  <div class="container-fluid">
    <div class="row">
      <nav id="sidebarMenu"
        class="col-md-3 col-lg-2 d-md-block bg-white border-end sticky-top vh-100 overflow-y-auto p-3">
        <div class="mb-4 ps-2">
          <h5 class="text-primary fw-bold mb-0">Vehicle ML</h5>
          <small class="text-muted">Regression Analysis</small>
        </div>
        <div id="data_exploration-nav" class="nav nav-pills flex-column">
          <a class="nav-link mb-1 active" href="#regression-analysis">Regression Analysis</a>
          <a class="nav-link mb-1" href="#evaluation-metrics">Evaluation Metrics</a>
          <a class="nav-link mb-1" href="{% url 'data_exploration' %}">Exploratory Data Analysis</a>
          <a class="nav-link mb-1" href="{% url 'classification_analysis' %}">Classification Analysis</a>
          <a class="nav-link mb-1" href="{% url 'clustering_analysis' %}">Clustering Analysis</a>
        </div>
      </nav>

      <main class="col-md-9 ms-sm-auto col-lg-10 px-md-4 py-4" data-bs-spy="scroll"
        data-bs-target="#data_exploration-nav" data-bs-smooth-scroll="true" tabindex="0">
        <header class="mb-4 border-bottom pb-3">
          <h1 class="fw-bold text-dark">Vehicle Insights Dashboard</h1>
          <p class="text-secondary mb-0">Real-time Machine Learning Inference</p>
        </header>

        <section id="regression-analysis" class="mb-5" style="padding-top: 20px;">
          <div class="row g-4">
            <div class="col-lg-7">
              <div class="card shadow-sm border-0 h-100">
                <div class="card-header bg-primary text-white py-3">
                  <h6 class="mb-0 fw-bold">Input Vehicle Specifications</h6>
                </div>
                <div class="card-body p-4">
                  <form method="POST" action="">
                    {% csrf_token %}
                    <div class="row g-3">
                      <div class="col-md-6">
                        <label class="form-label fw-semibold">Model Year</label>
                        <input type="number" name="year" class="form-control"
                          placeholder="e.g. 2022" required>
                      </div>
                      <div class="col-md-6">
                        <label class="form-label fw-semibold">Kilometers Driven</label>
                        <input type="number" step="any" name="km" class="form-control"
                          placeholder="e.g. 15000" required>
                      </div>
                      <div class="col-md-6">
                        <label class="form-label fw-semibold">Number of Seats</label>
                        <input type="number" name="seats" class="form-control"
                          placeholder="e.g. 5" required>
                      </div>
                      <div class="col-md-6">
                        <label class="form-label fw-semibold">Owner Income</label>
                        <input type="number" step="any" name="income" class="form-control"
                          placeholder="e.g. 50000" required>
                      </div>
                    </div>
                    <div class="col-12 mt-4">
                      <button type="submit" class="btn btn-primary w-100 fw-bold py-2">Predict
                        Market Price</button>
                    </div>
                  </form>
                </div>
              </div>
            </div>
          </div>
        </section>
      </main>
    </div>
  </div>
```

```

<div class="col-lg-5">
  <div class="card shadow-sm border-0 h-100 bg-white">
    <div class="card-header bg-dark text-white py-3">
      <h6 class="mb-0 fw-bold">Prediction Result</h6>
    </div>
    <div
      class="card-body d-flex flex-column align-items-center justify-content-center text-center p-4">
      {% if price %}
      <div class="mb-2 text-muted text-uppercase small fw-bold">Estimated Value</div>
      <h2 class="display-4 fw-bold text-success mb-2">${{ price|floatformat:2 }}</h2>
      {% else %}
      <div class="py-5">
        <div class="text-muted mb-3 opacity-25">
          Waiting...
        </div>
        <p class="text-secondary">Fill out the form to generate a price prediction.</p>
      </div>
      {% endif %}
    </div>
  </div>
</div>
</div>
</section>

<section id="evaluation_metrics" class="mb-5" style="padding-top: 20px;">
  <div class="card shadow-sm border-0">
    <div class="card-header bg-white border-bottom py-3">
      <span class="fw-bold">Evaluation Metrics</span>
    </div>
    <div class="card-body overflow-x-auto">
      Then, {{ evaluations.r2|safe }}% measures how well these selected features explain the
      variation in selling prices.
      {{ evaluations.comparison|safe }}
    </div>
  </div>
</section>
</main>
</div>
</div>

<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>

```

Create file within the app folder: templates/predictor/classification_analysis.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Vehicle ML Dashboard - Classification</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
</head>

<body class="bg-light">
  <div class="container-fluid">
    <div class="row">
      <nav id="sidebarMenu" class="col-md-3 col-lg-2 d-md-block bg-white border-end sticky-top vh-100 overflow-y-auto p-3">
        <div class="mb-4 ps-2">
          <h5 class="text-primary fw-bold mb-0">Vehicle ML</h5>
          <small class="text-muted">Classification Analysis</small>
        </div>

        <div id="data_exploration-nav" class="nav nav-pills flex-column">
          <a class="nav-link mb-1 active" href="#classification-analysis">Classification Analysis</a>
          <a class="nav-link mb-1" href="#evaluation_metrics">Evaluation Metrics</a>
          <a class="nav-link active mb-1" href="{% url 'data_exploration' %}">Exploratory Data Analysis</a>
          <a class="nav-link mb-1" href="{% url 'regression_analysis' %}">Regression Analysis</a>
          <a class="nav-link mb-1" href="{% url 'clustering_analysis' %}">Clustering Analysis</a>
        </div>
      </nav>

      <main class="col-md-9 ms-sm-auto col-lg-10 px-md-4 py-4">
        data-bs-spy="scroll"
        data-bs-target="#data_exploration-nav"
        data-bs-smooth-scroll="true"
        tabindex="0">

        <header class="mb-4 border-bottom pb-3">
          <h1 class="fw-bold text-dark">Vehicle Insights Dashboard</h1>
          <p class="text-secondary mb-0">Classification: Predicting Owner Income Category</p>
        </header>

        <section id="classification-analysis" class="mb-5" style="padding-top: 20px;">
          <div class="row g-4">
            <div class="col-lg-7">
              <div class="card shadow-sm border-0 h-100">
                <div class="card-header bg-primary text-white py-3">
                  <h6 class="mb-0 fw-bold">Input Vehicle Specifications</h6>
                </div>
                <div class="card-body p-4">
                  <form method="POST" action="">
                    {% csrf_token %}
                    <div class="row g-3">
                      <div class="col-md-6">
                        <label class="form-label fw-semibold">Model Year</label>
                        <input type="number" name="year" class="form-control" placeholder="e.g. 2022" required>
                      </div>
                      <div class="col-md-6">
                        <label class="form-label fw-semibold">Kilometers Driven</label>
                        <input type="number" step="any" name="km" class="form-control" placeholder="e.g. 15000" required>
                      </div>
                      <div class="col-md-6">
                        <label class="form-label fw-semibold">Number of Seats</label>
                        <input type="number" name="seats" class="form-control" placeholder="e.g. 5" required>
                      </div>
                      <div class="col-md-6">
                        <label class="form-label fw-semibold">Reference Income</label>
                        <input type="number" step="any" name="income" class="form-control" placeholder="e.g. 50000" required>
                      </div>
                    </div>
                    <div class="col-12 mt-4">
                      <button type="submit" class="btn btn-primary w-100 fw-bold py-2">Predict Income Category</button>
                    </div>
                  </form>
                </div>
              </div>
            </div>
          </div>
        </section>
      </main>
    </div>
  </div>
```

```

<div class="col-lg-5">
  <div class="card shadow-sm border-0 h-100 bg-white">
    <div class="card-header bg-dark text-white py-3">
      <h6 class="mb-0 fw-bold">Classification Result</h6>
    </div>
    <div class="card-body d-flex flex-column align-items-center justify-content-center text-center p-4">
      {% if prediction %}
        <div class="mb-2 text-muted text-uppercase small fw-bold">Predicted Category</div>
        <h2 class="display-4 fw-bold text-success mb-2">{{ prediction }}</h2>
      {% else %}
        <div class="py-5">
          <div class="text-muted mb-3 opacity-25">
            Waiting...
          </div>
          <p class="text-secondary">Fill out the form to classify the income level.</p>
        </div>
      {% endif %}
    </div>
  </div>
</div>
</div>
</div>
</section>

<section id="evaluation_metrics" class="mb-5" style="padding-top: 20px;">
  <div class="card shadow-sm border-0">
    <div class="card-header bg-white border-bottom py-3">
      <span class="fw-bold">Evaluation Metrics</span>
    </div>
    <div class="card-body overflow-x-auto">
      <p class="mb-3">The model has an accuracy of <strong>{{ evaluations.accuracy }}%</strong> in correctly categorizing the
income level based on historical vehicle data.</p>
      <div class="table-responsive">
        {{ evaluations.comparison|safe }}
      </div>
    </div>
  </div>
</div>
</section>
</main>
</div>
</div>

<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>

```

Create file within the app folder: templates/predictor/clustering_analysis.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Vehicle ML Dashboard</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
</head>

<body class="bg-light">
  <div class="container-fluid">
    <div class="row">

      <nav id="sidebarMenu"
        class="col-md-3 col-lg-2 d-md-block bg-white border-end sticky-top vh-100 overflow-y-auto p-3">
        <div class="mb-4 ps-2 border-bottom pb-3">
          <h5 class="text-primary fw-bold mb-0">Vehicle ML</h5>
          <small class="text-muted">Clustering Analysis</small>
        </div>
        <div id="data_exploration-nav" class="nav nav-pills flex-column">
          <a class="nav-link mb-1 active" href="#clustering-analysis">Clustering Analysis</a>
          <a class="nav-link mb-1" href="#evaluation-metrics">Evaluation Metrics</a>
          <a class="nav-link mb-1" href="#comparison-analysis">Comparison Analysis</a>
          <a class="nav-link mb-1" href="{% url 'data_exploration' %}">Exploratory Data Analysis</a>
          <a class="nav-link mb-1" href="{% url 'regression_analysis' %}">Regression Analysis</a>
          <a class="nav-link mb-1" href="{% url 'classification_analysis' %}">Classification Analysis</a>
        </div>
      </nav>

      <main class="col-md-9 ms-sm-auto col-lg-10 px-md-4 py-4" data-bs-spy="scroll"
        data-bs-target="#data_exploration-nav" data-bs-smooth-scroll="true" tabindex="0">

        <header class="mb-4 border-bottom pb-3 d-flex justify-content-between align-items-center">
          <div>
            <h1 class="fw-bold text-dark">Vehicle Insights Dashboard</h1>
            <p class="text-secondary mb-0">Regression & Clustering Analysis</p>
          </div>
        </header>

        <section id="clustering-analysis" class="mb-5 pt-3">
          <div class="row g-4">
            <div class="col-lg-7">
              <div class="card shadow-sm border-0 h-100">
                <div class="card-header bg-primary text-white py-3">
                  <h6 class="mb-0 fw-bold">Input Specifications</h6>
                </div>
                <div class="card-body p-4">
                  <form method="POST" action="">
                    {% csrf_token %}
                    <div class="row g-3">
                      <div class="col-md-6">
                        <label class="form-label fw-semibold small">Model Year</label>
                        <input type="number" name="year" class="form-control" placeholder="2022"
                          required>
                      </div>
                      <div class="col-md-6">
                        <label class="form-label fw-semibold small">Kilometers Driven</label>
                        <input type="number" step="any" name="km" class="form-control"
                          placeholder="15000" required>
                      </div>
                      <div class="col-md-6">
                        <label class="form-label fw-semibold small">Seats</label>
                        <input type="number" name="seats" class="form-control" placeholder="5"
                          required>
                      </div>
                      <div class="col-md-6">
                        <label class="form-label fw-semibold small">Owner Annual Income</label>
                        <input type="number" step="any" name="income" class="form-control"
                          placeholder="50000" required>
                      </div>
                    </div>
                    <div class="col-12 mt-4">
                      <button type="submit"
                        class="btn btn-primary w-100 fw-bold py-2 shadow-sm">Run Combined
                        Inference</button>
                    </div>
                  </form>
                </div>
              </div>
            </div>
          </div>
        </section>
      </main>
    </div>
  </div>
```

```

        </div>
    </div>
</form>
</div>
</div>
</div>

<div class="col-lg-5">
    <div class="card shadow-sm border-0 h-100 bg-white overflow-hidden">
        <div class="card-header bg-dark text-white py-3">
            <h6 class="mb-0 fw-bold">Dual-Model Result</h6>
        </div>
        <div
            class="card-body d-flex flex-column align-items-center justify-content-center text-center p-4">
            {% if price %}
            <div class="mb-4">
                <small class="text-muted text-uppercase fw-bold ls-wide">Estimated Value</small>
                <h2 class="display-4 fw-bold text-primary mb-0">${{ price|floatformat:2 }}</h2>
            </div>

            <hr class="w-75 my-3 opacity-10">

            <div class="mt-2">
                <small class="text-muted text-uppercase fw-bold ls-wide">Client Cluster</small>
                <div class="mt-2">
                    <span class="badge rounded-pill bg-success px-4 py-2 fs-6">
                        {{prediction|safe }}</span>

                </div>
            </div>
            {% else %}
            <div class="py-5">
                <div class="text-muted mb-3 opacity-25">Waiting...</div>
                <p class="text-secondary">
                    Awaiting input to generate prediction and customer profile.
                </p>
            </div>
            {% endif %}
        </div>
    </div>
</div>
</section>
<section id="evaluation-metrics" class="mb-5 pt-3">
    <div class="card shadow-sm border-0">
        <div
            class="card-header bg-white border-bottom py-3 d-flex justify-content-between align-items-center">
            <span class="fw-bold text-dark">Evaluation Metrics</span>
            <span class="badge bg-light text-primary border border-primary">
                Silhouette Score: {{evaluations.silhouette|safe }}
            </span>
        </div>
        <div class="card-body">
            <div class="table-responsive">
                {{ evaluations.summary|safe }}
            </div>
        </div>
    </div>
</section>
<section id="comparison-analysis" class="mb-5 pt-3">
    <div class="card shadow-sm border-0">
        <div class="card-header bg-white border-bottom py-3">
            <span class="fw-bold">Comparison Analysis</span>
        </div>
        <div class="card-body p-0">
            <div class="table-responsive mt-3">
                {{ evaluations.comparison|safe }}
            </div>
        </div>
    </div>
</section>
</main>
</div>
</div>
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>

```

16. Update the app urls.py

```
path("regression_analysis",views.regression_analysis, name="regression_analysis"),  
path("classification_analysis",views.classification_analysis,name="classification_analysis"),  
path("clustering_analysis",views.clustering_analysis, name="clustering_analysis"),
```

17. Running the Server

Run the Django development server.

```
python manage.py runserver
```

18. Testing the System

Open browser: `http://127.0.0.1:8000`

Enter vehicle data and check the predicted price.

19. Exercise

Students can add dashboards using Plotly.

- a) On exploratory data analysis, display the Rwanda map with names and districts boundaries, number of vehicle clients in each district. **(20 marks)**
- b) The Silhouette Score is 0.68,
 - Calculate the *coefficient of variation* and display it along with Silhouette Score. **(5 marks)**
 - Refine the model Silhouette Score to have the Silhouette Score above 0.9**(5marks)**